FYS4480/9480 AUG 27

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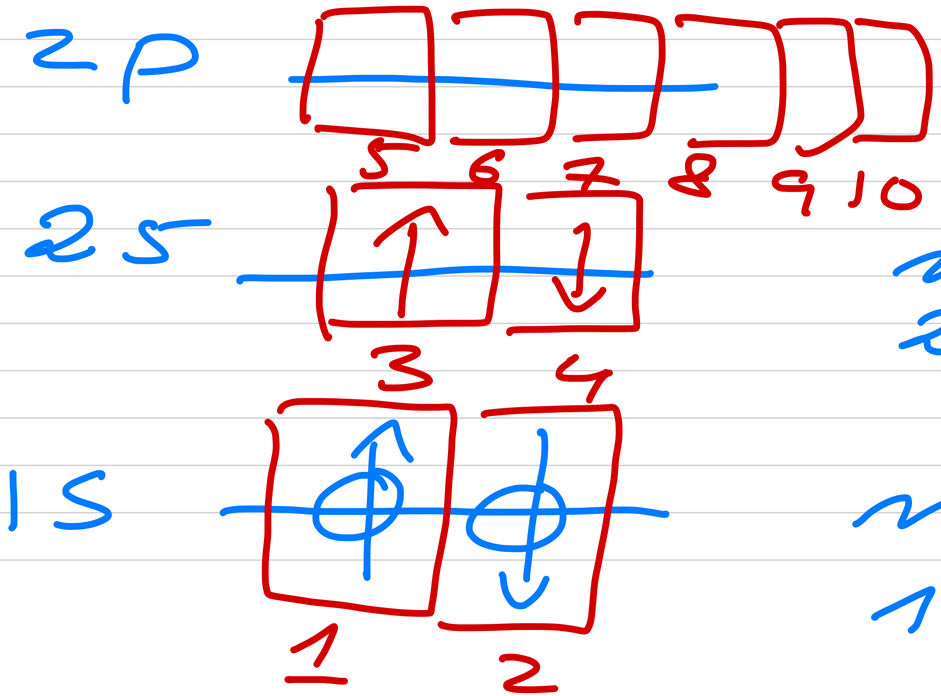
Last week: $N = 2$

$$\hat{h}_0 \varphi_1(x) = \varepsilon_1 \varphi_1(x)$$

$$\hat{h}_0 \varphi_2(x) = \varepsilon_2 \varphi_2(x)$$

Atomic helium



$n \quad l \quad m_l$

$2 \quad 1 \quad -1, 0, 1$

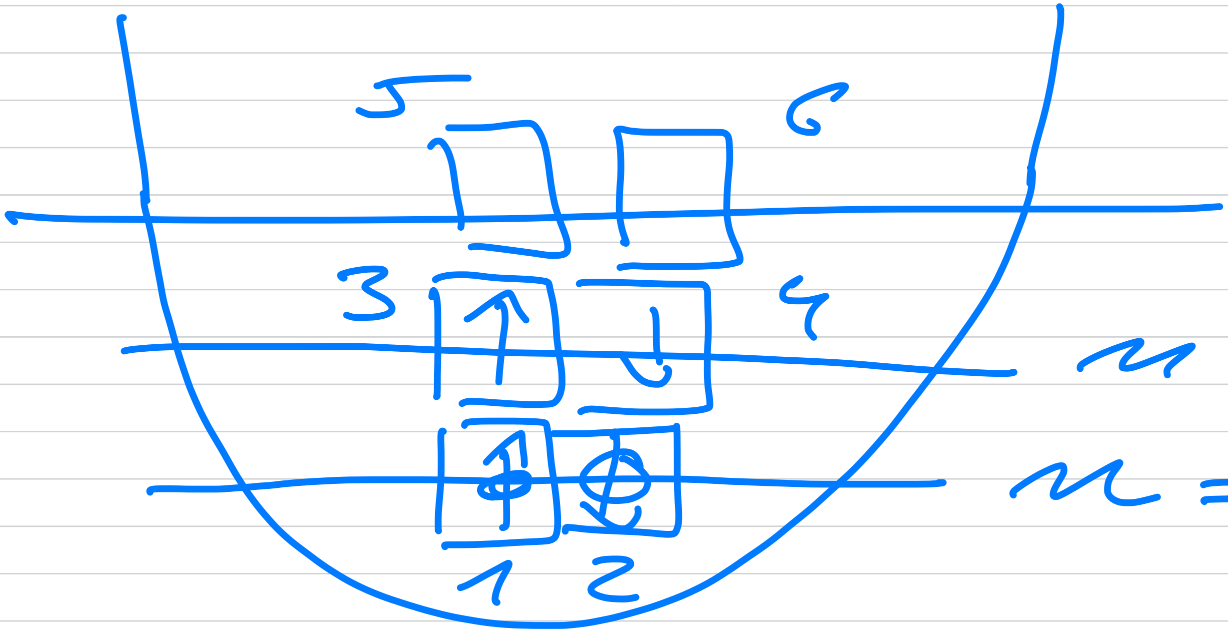
$n \quad l \quad m_l$

$2 \quad 0 \quad 0$

$n \quad l \quad m_l$

$1 \quad 0 \quad 0$

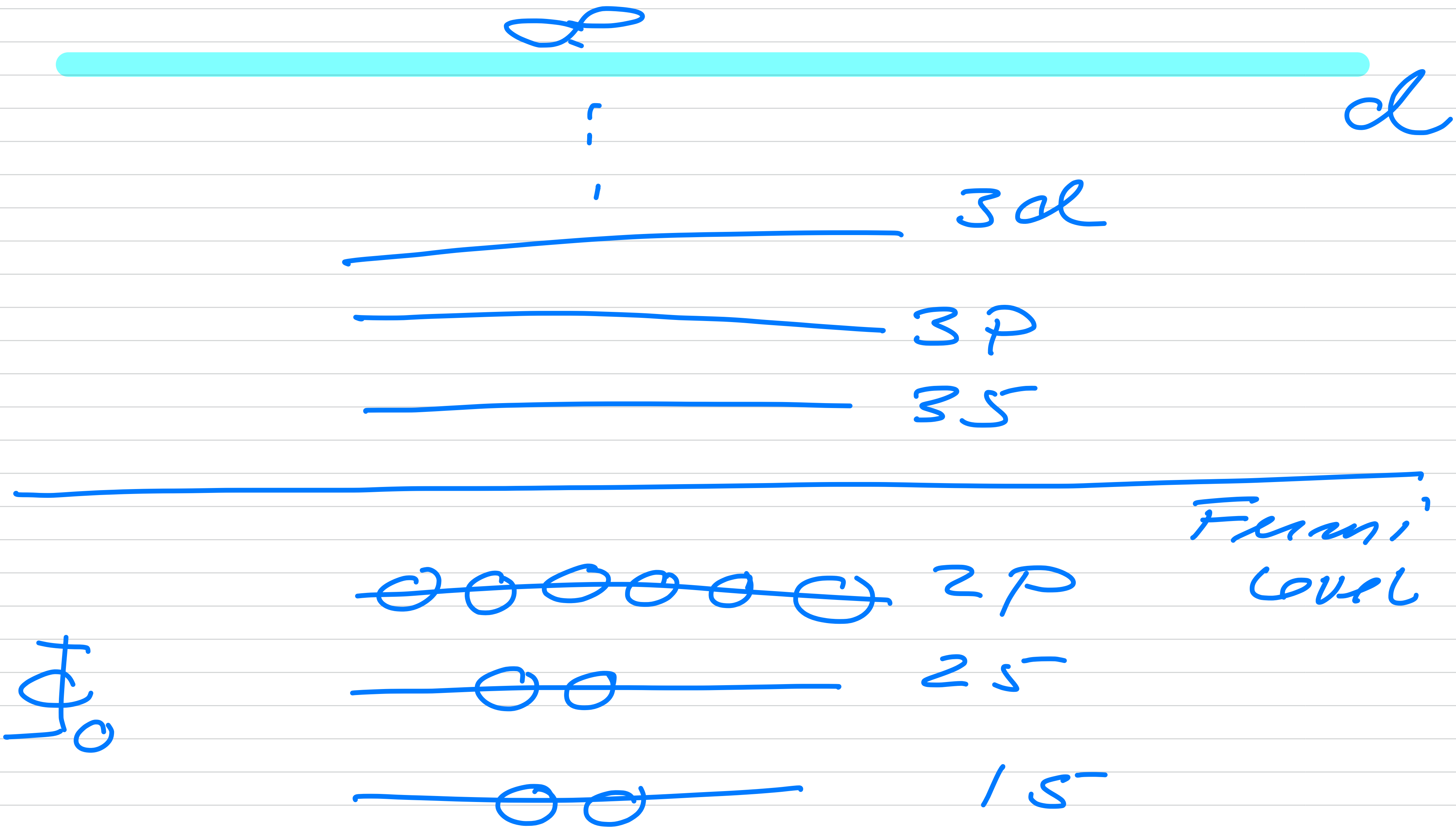
1-dim HO



$n = 1 \quad \varepsilon_1$

$n = 0 \quad \varepsilon_0$

Fermi level Atomic Neon



states (configurations)

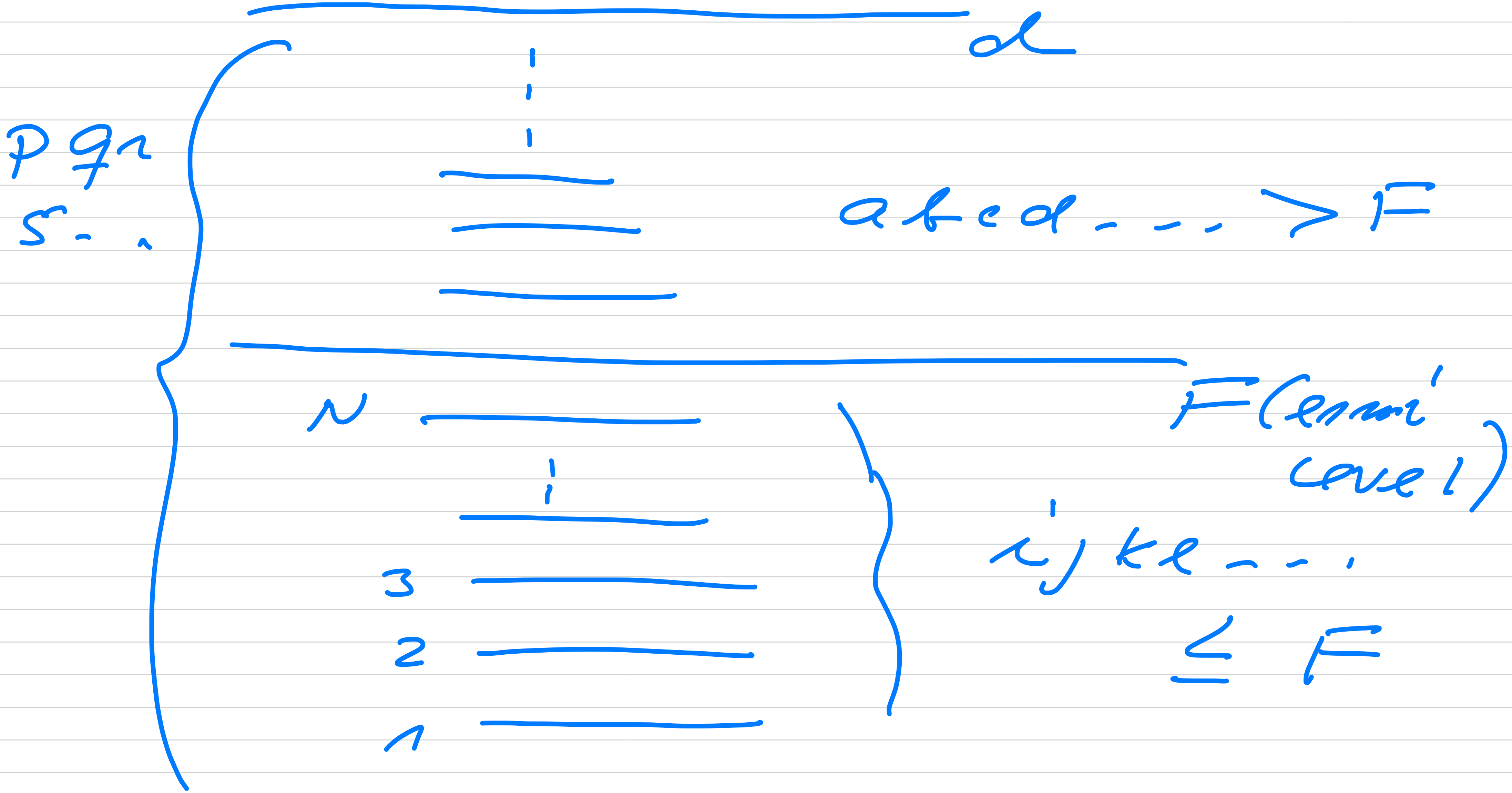
n - sp-states

N - particles

$$\binom{n}{N} = \frac{n!}{(n-N)!\,N!}$$

$n \geq N$

Compact notation



Helium ansatz for Φ_0

$$\mathcal{H}\,\Phi_0 \neq E_0\,\Phi_0$$

$$\mathcal{H}\,\Psi_0 = E_0\,\Psi_0$$

$$\hat{\mathcal{H}}_0\,\Phi_0 = \varepsilon_0\,\Phi_0$$

$$\Phi_0(x_1, x_2) = \frac{1}{\sqrt{2!}}\Big(\varphi_1(x_1)\,\varphi_2(x_2) - \varphi_1(x_2)\,\varphi_2(x_1)\Big)$$

$$= \frac{1}{\sqrt{2!}}\begin{vmatrix} \varphi_1(x_1) & \varphi_1(x_2) \\ \varphi_2(x_1) & \varphi_2(x_2) \end{vmatrix}$$

$$\langle \Phi_0 | \hat{\mathcal{H}} | \Phi_0 \rangle$$

$$\langle 2 | \hat{h}_0 | 2 \rangle$$

$$\int dx\, \varphi_2^*(x)\, \hat{h}_0\, \varphi_2(x)$$

$$= \quad \varepsilon_1 + \varepsilon_2 + \langle 12 | v | 12 \rangle - \langle 12 | v | 21 \rangle$$

Direct $\qquad\qquad$ exchange

$$\int dx_1 \int dx_2\, \varphi_1^*(x_1)\, \varphi_2^*(x_2)\, v(x_1 x_2)\, \varphi_1(x_1)\, \varphi_2(x_2)$$

$N > 2$

$$\langle \Phi_0 | \mathcal{H} | \Phi_0 \rangle = ?$$

ansatz for Ψ_0 is Φ_0

$$\Phi_0(x_1\, x_2 \ldots x_N) =$$

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_1(x_1) & \cdots & \varphi_1(x_N) \\ \varphi_2(x_1) & & \varphi_2(x_N) \\ \vdots & & \vdots \\ \varphi_N(x_1) & & \varphi_N(x_N) \end{vmatrix}$$

$$\Psi_0 = \sum_{j=0}^{\infty} c_{0j} \Phi_j$$

Leibniz rep. for a deter-minant

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & & & \\ a_{31} & & & \\ \vdots & & & \\ a_{N1} & \dots & \dots & a_{NN} \end{bmatrix}$$

$$\det(A) = \sum_{P \in S_N} \underbrace{\operatorname{sgn}(P)}_{(-1)^P} \prod_{i=1}^{N} a_{i(Pi)}$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}\underbrace{a_{21}}_{1 \leftrightarrow 2}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & & \\ a_{31} & - & -\, a_{33} \end{vmatrix} \qquad 3! = 6$$

$$\underset{(1\,2\,3)}{a_{11}a_{22}a_{33}} - \underset{(1\,3\,2)}{a_{11}a_{23}a_{32}} +$$

$$+ \underset{(2\,1\,3)}{a_{12}}(a_{21}a_{33} - a_{31}a_{23}) - \ldots$$

antisym operator

$$\hat{A} = \frac{1}{N!} \sum_{p \in S_N} (-)^p \hat{P}$$

$$\hat{A}^2 = \hat{A} \quad ; \quad \hat{A}^\dagger = A$$

$N = 2$ $\qquad a_{11} a_{22} - \underbrace{a_{12} a_{21}}$

$$\hat{A}_2 = \frac{1}{2}\left(1 - \underline{\hat{P}_2}\right)$$

$$\hat{A}_2^2 = \frac{1}{4}\left(1 - 2P_2 + \underbrace{\hat{P}_2^2}_{+\underline{1}}\right)$$

$$= \frac{1}{2}\left(1 - \hat{P}_2\right) = \hat{A}$$

$$[\hat{A}, \hat{H}] = [\hat{A}, \hat{H}_0] + [\hat{A}, \hat{H}_I]$$

$$= 0$$

$$\Phi_0 = \frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_1(x_1) & \cdots & \varphi_1(x_N) \\ \varphi_2(x_1) & & \vdots \\ \vdots & & \vdots \\ \varphi_N(x_1) & & \varphi_N(x_N) \end{vmatrix}$$

$$= \sqrt{N!}\, \hat{A}\, \Phi_H (x_1, x_2, \ldots x_N)$$

Hartree - function

$$\Phi_H (x_1 x_2 \ldots x_N) =$$

$$\varphi_1(x_1)\, \varphi_2(x_2) \ldots \varphi_N(x_N)$$

$$(a_{11} \quad a_{22} \quad \ldots \quad a_{NN})$$

$$- a_{12} a_{21}$$

$$\langle \Phi_0 | \mathcal{H}_c | \Phi_0 \rangle = ?$$

$$\langle \Phi_0 | \hat{H}_0 | \Phi_0 \rangle$$

$$= N! \int d\tau \, \Phi_H^* \hat{A}^\dagger \hat{H}_0 \hat{A} \Phi_H$$

$$\left(\begin{array}{l} d\tau = dx_1 dx_2 \ldots \quad dx_N \\ x_i = (\vec{r}_i, \sigma_i) \end{array} \right)$$

$$= N! \int d\tau \, \Phi_H^* \hat{H}_0 \hat{A} \Phi_H$$

$$\hat{A}^\dagger \hat{H}_0 = \hat{H}_0 \hat{A}^\dagger \qquad \hat{A}^2 = \hat{A}$$

$$= \int dx\, \Phi_H^* \hat{H}_0 \sum_{P \in S_N} (-)^P \hat{P}\, \Phi_H$$

0 - permutation

$$\int dx_1 \int dx_2 \cdots \int dx_N\, \varphi_1^*(x_1) \cdots \varphi_N^*(x_N)$$

$$\times \left(\hat{h}_0(x_1) + \hat{h}_0(x_2) + \cdots \hat{h}_0(x_N) \right)$$

$$\varphi_1(x_1)\, \varphi_2(x_2) \cdots \varphi_N(x_N)$$

$$= \sum_{i=1}^{N} \langle i | \hat{h}_0 | i \rangle = \sum_{i=1}^{N} \varepsilon_i$$

$$\int dx_i\, \varphi_i^*(x_i)\, \hat{h}_0(x_i)\, \varphi_i(x_i)$$

$$\hat{h}_0(x)\, \varphi_i(x) = \varepsilon_i\, \varphi_i(x)$$

$$\int dx\, \varphi_i^*(x)\, \varphi_j(x) = \delta_{ij}$$

1 - permutations $(1 \leftrightarrow 2)$
(and applies to all other)

$$\int dx_1 \int dx_2 \int \cdots \int dx_N \times$$

$$\varphi_1^*(x_1)\,\varphi_2^*(x_2) \cdots \varphi_N^*(x_N)$$

$$\times \left(\hat{h}_0(x_1) + \hat{h}_0(x_2) + \cdots \hat{h}_0(x_N) \right)$$

$$\times\, \varphi_1(x_2)\,\varphi_2(x_1)\,\varphi_3(x_3) \cdots \varphi_N(x_N)$$

$$\underbrace{\boxed{\int dx_1\, \varphi_1^*(x_1)\,\hat{h}_0(x_1)\,\varphi_2(x_1)}}_{}\overset{=0}{} \int dx_2$$

$$\underbrace{\varphi_2^*(x_2)\,\varphi_1(x_2)}_{=0} \int dx_3\, \varphi_3^*(x_3)\,\varphi_3(x_3)$$

$$\cdots \int \cdots$$

$$\hat{H}_0 \Phi_0 = \mathcal{E}_0 \Phi_0$$

$$\mathcal{E}_0 = \sum_{i=1}^{N} \varepsilon_i$$

$$\boxed{\hat{H}_0 \Phi_i = \mathcal{E}_i \Phi_i}$$

$$\langle \Phi_0 | H_I | \Phi_0 \rangle$$

$$H_I = \sum_{i<j} v(x_i, x_j) = \sum_{i<j} v(|x_i - x_j|)$$

$$\langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle$$

$$= N! \int d\tau \, \Phi_H^* \hat{H}_I \hat{A} \Phi_H$$

$$= \int d\tau \, \Phi_H^* \hat{H}_I \sum_{P \in S_N} (-)^P \hat{P} \Phi_H$$

0-permutation

$$\int dx_1 \int dx_2 \cdots \int dx_N$$

$$\times\ \varphi_1^*(x_1)\,\varphi_2^*(x_2) \cdots \varphi_N^*(x_N)$$

$$\times \sum_{i'<j'} v(x_{i'}, x_{j'})$$

$$\times\ \varphi_1(x_1)\,\varphi_2(x_2) \cdots \varphi_N(x_N)$$

$$= \sum_{i<j} \int dx_{i'} \int dx_{j'}\ \varphi_i^*(x_{i'})\,\varphi_j^*(x_{j'})\ v(x_{i'}, x_{j'})\ \varphi_i(x_{i'})\,\varphi_j(x_{j'})$$

$$= \sum_{i<j} \langle ij|v|ij\rangle$$ Direct term

1 - permutation

$x_1 \Longleftrightarrow x_2 \qquad (-1)^1$

$$\int dx_1\, dx_2 \,--\, dx_N\, \varphi_1^*(x_1)\, \varphi_2^*(x_2)$$

$$--\, \varphi_N^*(x_N)\, v(x_1, x_2)\, \varphi_1(x_2)\, \varphi_2(x_1)$$

$$\times \int dx_3\, \varphi_3^*(x_3)\, \varphi_3(x_3) \,--\, \int dx_N\, \varphi_N^* \varphi_N$$

$$\int dx_2\, \varphi_2^*(x_2)\, \varphi_1(x_2) \,--$$

$$+ \int --- \qquad v(x_1, x_3) \,---$$

$$+ \int --- \qquad v(x_1, x_4) \,---$$

$$= - \int dx_1 \, dx_2 \, \varphi_1^*(x_1) \, \varphi_2^*(x_2)$$

$$v(x_1 x_2) \, \varphi_1(x_2) \, \varphi_2(x_1)$$

$$= - \langle 12 | v | 21 \rangle$$

$x_1 \Leftrightarrow x_3$

$$= - \langle 13 | v | 31 \rangle$$

$$\langle \Phi_0 | \mathcal{U}_I | \Phi_0 \rangle =$$

$$\sum_{i<j} \left\{ \langle ij | v | ij \rangle - \langle ij | v | ji \rangle \right\}$$